

Derivation of Stiffness and Mass Matrices for a 2-Node Euler–Bernoulli Beam Element

1. Degrees of Freedom

For a 2-node beam element:

$$\mathbf{q} = [w_1 \quad \theta_1 \quad w_2 \quad \theta_2]^T$$

2. Cubic Hermite Shape Functions

Let $\xi = x/L$.

$$\begin{aligned} N_1 &= 1 - 3\xi^2 + 2\xi^3, \\ N_2 &= L(\xi - 2\xi^2 + \xi^3), \\ N_3 &= 3\xi^2 - 2\xi^3, \\ N_4 &= L(-\xi^2 + \xi^3) \end{aligned}$$

Then

$$w(x) = \mathbf{N}(x)\mathbf{q}$$

with

$$\mathbf{N} = [N_1 \ N_2 \ N_3 \ N_4].$$

3. Stiffness Matrix

Strain energy:

$$U = \frac{1}{2} \int_0^L EI \left(\frac{d^2 w}{dx^2} \right)^2 dx$$

Define

$$\mathbf{B} = \frac{d^2 \mathbf{N}}{dx^2}$$

Then

$$U = \frac{1}{2} \mathbf{q}^T \left(\int_0^L EI \mathbf{B}^T \mathbf{B} dx \right) \mathbf{q}$$

Hence

$$\mathbf{K} = \int_0^L EI \mathbf{B}^T \mathbf{B} dx$$

Result:

$$\mathbf{K} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

4. Mass Matrix

Kinetic energy:

$$T = \frac{1}{2} \int_0^L \rho A (\dot{w})^2 dx$$

Using $\dot{w} = \mathbf{N}\dot{\mathbf{q}}$:

$$T = \frac{1}{2} \dot{\mathbf{q}}^T \left(\int_0^L \rho A \mathbf{N}^T \mathbf{N} dx \right) \dot{\mathbf{q}}$$

Hence

$$\mathbf{M} = \int_0^L \rho A \mathbf{N}^T \mathbf{N} dx$$

Result:

$$\mathbf{M} = \frac{\rho AL}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$